Paper 7

Acceleration, Momentum, and Geometric Consolidation

Paper 7 of the Series

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ABSTRACT

Paper 6 established that uniform motion is a self-sustaining directional offset in a configuration's ceaseless ledger, costing nothing to maintain, and that inertia is the workload of re-cutting that offset into a new one. It left the quantitative accounting of that workload — what a change of motion costs, what a moving configuration carries, and where the delivered energy goes — to this paper. Paper 7 supplies it. Acceleration is the continuous re-cutting of a configuration's offset: where uniform motion is a standing offset that demands no input, acceleration is the tick-by-tick rewriting of that offset, which can only proceed while geometry is actively delivered, because re-cutting a ledger is added geometry and added geometry must come from somewhere. This is why a force is needed to accelerate and not to coast — the force is the delivery of the geometry the re-cut consumes. Momentum is the standing offset itself, read as a conserved transferable quantity: the accumulated directional geometry packed into a configuration's ledger, indivisible and quantized (Paper 6, Section 4), which transfers whole across junctions and so is conserved exactly in every interaction. Geometric consolidation is the mechanism that makes the two cohere: when delivered geometry is re-cut into a configuration during acceleration, it does not sit as a passenger and does not leak back out when delivery stops; it is structurally absorbed into the standing ledger as a permanent offset, so that the configuration continues at its new motion with no further input. The paper grounds all three in two worked demonstrations — the chemical rocket, which generates directional geometry internally by creation events and expels the waste to drive a forward wedge into its own structure, and the magnetic railway, which projects a traveling wave of directional geometry into the vehicle as an external write-head with near-zero leakage. It establishes that acceleration, unlike uniform motion, is absolute and self-detectable, because the re-cutting is a real ongoing workload the configuration performs on itself — the place where Paper 6's hidden offset surfaces. Because nothing is ever at rest — every configuration is always travelling, having stepped away from the photon baseline — acceleration is never the waking of a still object but the injection of brand-new creation geometry into an already-moving circuit, re-cutting its heading. A third demonstration, the slight deceleration of deep-space probes, is planted as the seed of the critique companion Paper 7.5. The composite case, in which the momenta of many bound sub-configurations consolidate into one momentum for the whole, is the bridge to Paper 16 and is marked here, not developed. Throughout, the accounting is closed and local under what this paper names Strict Generative Accounting (defined in Appendix D): conservation here is the stronger statement that the ledger balances on every tick even when balancing requires writing new coordinate geometry, not only relocating old — never freely, only as the mandatory cost of resolving a destabilized junction. The same accounting, summed across countless local conflicts and the outward deposit of unresolved scan workload by the searching states, is the mechanical seed of cosmic expansion, developed in Paper 14 and only pointed to here.

1. Purpose: From the Standing Offset to Its Change

Paper 6 fixed what motion is: a configuration runs a ceaseless ledger, and uniform motion is that ledger carrying a self-sustaining directional offset, stepped along a vector tick after tick, costing nothing to maintain because a conserved offset in a continuous medium does not erode. That paper named inertia — the resistance to a change of motion — as the workload of re-cutting an established cycle into a new one, and then, by its own contract, handed the quantitative accounting of that workload forward. This paper takes it up.

Three questions are owed, and they are the paper's three subjects. First, what is acceleration, mechanically — what is actually happening when a configuration's motion changes, and why does that, unlike uniform motion, require a continuous force? Second, what is momentum — what does a moving configuration physically carry, such that it is conserved exactly across every collision and transfer? Third, what is the mechanism, named here geometric consolidation, by which delivered geometry becomes permanent standing motion, so that a configuration accelerated and then released continues at its new rate with nothing further supplied? The three are one process seen at three moments — the delivery, the carrying, and the keeping — and the paper treats them in that order.

The contract of this paper. This paper *establishes* the mechanical identity of acceleration as the continuous re-cutting of a configuration's offset, of momentum as the conserved standing offset, and of consolidation as the permanence of delivered geometry; and it shows the conservation of momentum to be the conservation of total offset across a junction. It *assumes* from Papers 4, 5, 6, and 6.5 the photon baseline, energy as added geometry, the Closure Principle, the ceaseless ledger, the balanced and offset cycles, the partner-scan architecture, kinetic energy as delivered geometry (Paper 6.5), work as the phase-lock overload that writes an offset step, and the processing-budget conservation of the moving configuration. It does *not* derive the quantitative law of relativistic mass increase or the limiting behavior near c — that accounting belongs to Paper 8 — nor the law of time dilation (Paper 13), nor gravity (Paper 10). The composite consolidation of many momenta into one is marked for Paper 16. What this paper fixes is the mechanics of changing a state of motion and the conserved quantity that change deposits.

2. Acceleration as Continuous Re-Cutting

Figure 1. Holding an Offset versus Re-Cutting It: Why Coasting Is Free and Accelerating Is Not

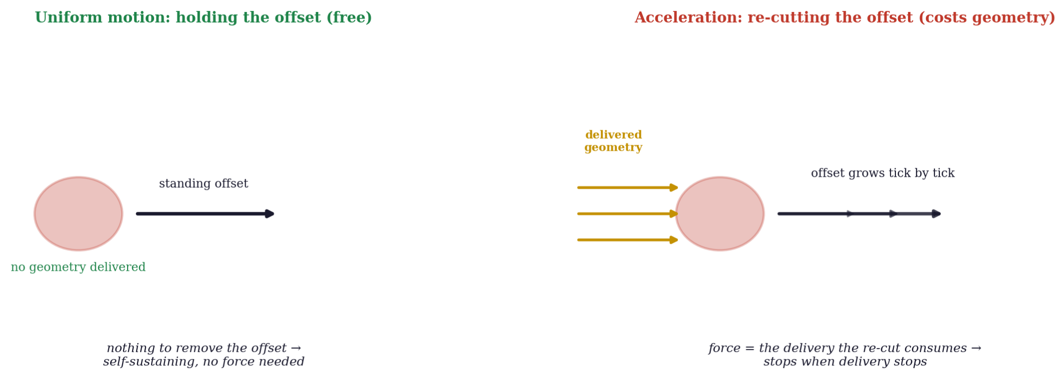

Figure 1. The distinction the paper turns on: holding a standing offset is free (nothing can remove it), while re-cutting it into a new offset consumes delivered geometry tick by tick — which is why coasting needs no force and accelerating does.

The distinction between uniform motion and acceleration is, in this framework, the distinction between holding an offset and changing one — and it is sharp, because the two have entirely different relationships to delivered geometry.

A configuration in uniform motion holds a standing offset. Paper 6 established why this needs no input: the offset is a quantized, indivisible ledger entry that cannot bleed into the surrounding baseline, because the medium outside the configuration runs pure baseline and has no slots to absorb a partial vector. The offset is self-sustaining not because something maintains it but because nothing can remove it. Holding a velocity is free.

Acceleration is not holding an offset; it is *re-cutting* it — replacing the configuration's current offset with a different one, and then a different one again, tick after tick. And re-cutting a ledger is not free, because rewriting geometry is added geometry, which is energy (Paper 4); this re-cut is Active Ledger Overwriting (Appendix D). Each increment of change demands an increment of delivered geometry to pay for the new offset step. This is the entire reason a force is required to accelerate and not to coast: the force is precisely the delivery of the geometry that the re-cutting consumes. A clarification the term *force* demands: only forces that change a configuration's motion are deliveries of geometry in this sense. Forces that merely constrain motion without changing it — a floor supporting a weight, static friction holding a block — deliver no new geometry; they are ledger boundary conditions, geometric constraints that forbid a path, not active re-cuts of an offset. The framework reserves *delivery* for the active case; the constraint case writes nothing new.

That re-cutting is not instantaneous across a composite body. When un-anchored creation geometry is delivered to a composite assembly, it creates a local coordinate overload at the boundary junction where it arrives. Because the assembly is a composite of many junctions, the overload does not process across the whole structure in a single tick; it propagates, junction to junction, as a physical compression wave — a conduction pump — rewriting the manifold scan at the processing rate of the medium, which is c. The trailing junctions, closest to the delivery, are already running the new directional offset while the leading junctions are still executing the previous cycle, and this frame-by-frame offset between junctions, travelling through the body at c, is the propagation of the re-cut. The structural deformation observed during hard acceleration — the compression of a chassis, the squeeze of a frame — is not a secondary mechanical side effect sitting outside the framework's account; it is the direct physical manifestation of this processing-rate lag, the rewrite travelling through the hardware junction by junction. The object squeezes because the code rewrite is in transit across it. The propagation speed of that lag is c: the front and back of a body do not move at once, because the rewrite is bounded by the same processing rate that bounds everything else.

The propagation is at c, not at some slower mechanical wave speed like sound in a solid, and this follows from the framework's most basic commitment: the monolithic processing rate. The reconciliation with the observed speed of sound is direct and does not require conceding a second, slower mechanism. When one end of a steel bar is struck, the ledger rewrite does propagate through the local junctions at c. But steel is a massive, multi-junction configuration tilted far from the photon baseline, and its fixed net power of c is heavily split — most of it committed to processing its own dense internal matrix, only a share left for passing a workload across the macroscopic body. What the world calls the speed of sound is not a separate, slower wave or a second physical rule; it is the observed macroscopic rate at which a heavily mass-loaded, tilted configuration can pass a workload through its internal matrix while running under a fixed net power of c. The ledger update never slows — the configuration's mass-load is what drags down the net translation of that update across the bulk. One rate, c, split by tilt; the acoustic speed is its macroscopic shadow in dense matter, not a rival to it. The medium does not have several speeds; it has one clock rate, c, and c is not introduced here as "the speed of light" borrowed from electromagnetism but as the universal processing ceiling of the medium, of which light's speed is one manifestation. The reconciliation with mass is the budget split inherited from Paper 6: a baseline photon commits its entire processing budget to spatial translation and so moves at c, while a massive configuration diverts a fixed share of its budget into running the internal ledger loop and the two-sweep identity audit that hold its departure together (Papers 2, 6). This split is established in the earlier papers and invoked here, not introduced. The massive configuration looks slower macroscopically, but it has not slowed in the hardware — its budget is split, translation plus internal scanning, and the two always sum to the same fixed total, c. Even a configuration "at rest" in a laboratory is scanning its neighbors at c internally.

This is why a structural rewrite propagates at c rather than at an acoustic speed. The propagation of a ledger re-cut is not atoms mechanically bumping their neighbors, which would indeed be slow; it is the medium recalculating coordinate steps at the junction interface, and that recalculation is executed by the same medium, on the same clock, that carries a free photon. The lag between trailing and leading junctions is processing latency — the tick-by-tick time for the medium to update the axis ratios across a saturated junction array — not a sound wave in matter. The framework therefore treats light propagation, inertial resistance, and structural-rewrite distribution as three manifestations of one synchronized clock rate. This is stated as the framework's foundational working commitment rather than derived from electromagnetism here; the demonstration that the medium's processing rate and the electromagnetic c are the same constant rests on the baseline mechanics of Paper 4, and is invoked, not re-proved, in this paper. When delivery is steady, the re-cutting is steady, and the configuration's offset grows or turns at a steady rate — a constant acceleration is a constant rate of geometry delivery being re-cut into the ledger. When delivery stops, the re-cutting stops, the current offset simply stands, and the configuration coasts at whatever motion the last completed re-cut left it in. That coasting is free does not mean a coasting configuration is unaffected by the medium: it is always interacting with the ambient baseline load that Paper 10 treats as gravity. But that interaction does not erode the standing offset — it curves the path the offset is carried along. The offset itself is held for free; the path it traces through the medium is shaped by the medium's geometry, which is a separate accounting handed to Paper 10.

This recovers the classical facts without their unexplained primitives. Newton's first law — coasting needs no force — is the standing offset's self-sustenance (Paper 6). The need for a force to accelerate is the cost of re-cutting. And the proportionality of that cost to mass, established qualitatively in Paper 6 as the linear workload of re-cutting more units of standing departure, is here the same fact read dynamically: a larger configuration has more ledger to re-cut per increment of offset change, so a given delivery of geometry produces a smaller change of motion. The framework does not assert the relation between force, mass, and change of motion; it exhibits the relation as the bookkeeping of delivered geometry re-cut against the count of units that must be re-cut.

Acceleration is absolute, and this is where the hidden offset surfaces. Because the re-cutting is a physical process occurring within the configuration itself rather than a relation between observers, it remains detectable even when no external reference is available. Paper 6 carried a debt: the standing offset of uniform motion is real but undetectable from inside the frame, which raised the charge of an unfalsifiable hidden variable, answered with the promise that the offset surfaces under acceleration. Here it surfaces. Re-cutting the ledger is a real workload the configuration performs, and unlike the standing offset — which is shared by every co-moving instrument and so cancels from every internal comparison — the *act of re-cutting* is not shared by a co-moving frame, because there is no inertial co-moving frame for an accelerating configuration; its own ledger is being actively rewritten in a way a uniformly moving reference is not. The configuration therefore detects its own acceleration directly, as the internal workload of its own continuous re-cutting, with no need to look outside. A uniformly accelerating frame is no exception: it is a frame whose ledger is being continuously rewritten, which is exactly what disqualifies it from being inertial. There is no co-moving inertial frame for an accelerating configuration, because the acceleration is a fact about the configuration's ledger, not a choice of the observer's coordinates. This is the framework's account of why acceleration is absolute while uniform motion is relative: uniform motion is a held offset that hides in shared cancellation, and acceleration is an ongoing re-cut that no co-moving reference shares. The asymmetry Paper 6 promised would surface is the asymmetry between holding and rewriting.

3. Momentum as the Conserved Standing Offset

Figure 2. Conservation of Momentum: Total Offset Transfers Whole Across the Junction

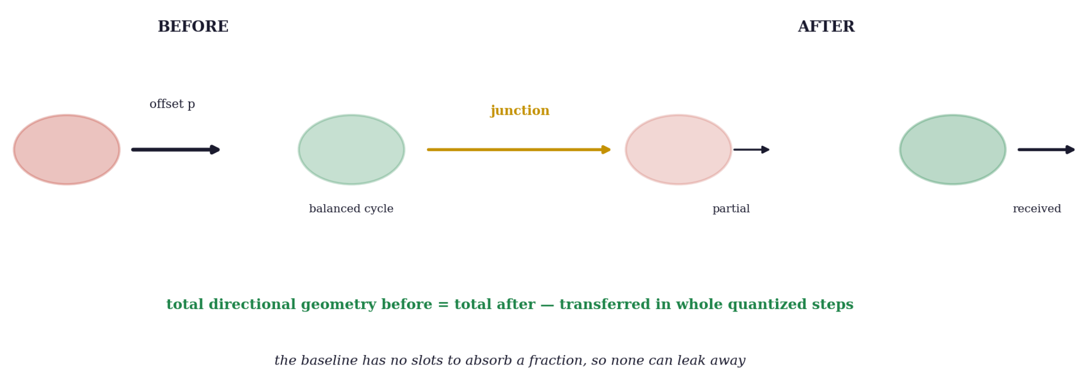

Figure 2. Conservation of momentum as conservation of total offset: directional geometry transfers whole, in quantized steps, across the junction, and cannot leak into a baseline that has no slots to hold a fraction of it.

If acceleration is the re-cutting and the standing offset is its result, momentum is that result regarded as a quantity — what the configuration carries, counted.

Momentum, in this framework, is the accumulated directional geometry packed into a configuration's ledger: the total of the offset that has been successfully re-cut into the standing cycle, pointing along the vector of motion. A word is owed here on what *geometry* means, because the term carries three distinct senses across the framework and momentum is only one of them. *Primitive geometry* is the X, Y, O values themselves, always conserved at three of each per triangle (the non-tunable hardware of Paper 2). *Offset geometry* is the directional imbalance of those values — the standing offset, which is momentum — held as the ratio of stepped updates across the three native axes. *Creation geometry* is new offset geometry introduced during a structural conflict, generated under load rather than conjured from nothing. Conservation of primitive geometry is absolute; conservation of offset geometry is exact at the ledger level; creation geometry is the mechanism by which offset geometry is introduced without violating primitive conservation. The full hierarchy is set out in Appendix E; momentum, throughout this paper, is offset geometry. It is not a property the configuration is assigned by a reference frame; it is a physical pile of directional geometry held in the active internal state, the same offset that steps the configuration across the medium each tick, regarded now as a stored amount rather than as a rate of stepping. The more geometry delivered and consolidated, the larger the standing offset, and the faster the configuration's own ledger requires it to travel.

This makes the conservation of momentum a structural necessity rather than an empirical law to be posited. The offset is a quantized, indivisible ledger entry (Paper 6): it cannot be partially destroyed or partially created, because the medium has no slots for fractional vectors. In any interaction, therefore, the total directional geometry present before the interaction must be present after it — it can be transferred, in whole quantized steps, from one configuration's ledger into another's, but it cannot leak away into the baseline (which cannot count it) or appear from the baseline (which has none to give). Conservation of momentum is conservation of total offset across the junction: the books on directional geometry must balance because directional geometry is indivisible and the baseline cannot hold any of it. This conservation and the generation of new geometry under Strict Generative Accounting are not in tension. Generation adds coordinate volume. It does not add directional offset. Momentum is conserved exactly; generation is how the ledger stays balanced when transfer alone cannot. To be precise about the word: when momentum transfers *whole* between ledgers, what transfers whole is the directional offset quantity — the indivisible quantized ledger entry of Paper 6, not the whole configuration and not its entire ledger — and that offset quantity cannot be created from the baseline or lost into it. The geometry generated under a high-torque conflict is new coordinate volume written to resolve a destabilized junction, never free directional momentum conjured from nothing. A collision conserves the total directional offset exactly; what generation adds is the coordinate volume the medium must write to keep that exact accounting possible under load. The momentum ledger balances; the generative ledger is what lets it balance when transfer alone cannot. The collision mechanics already established for the kinetic case (Paper 6.5, Appendix E) are the same mechanics here — excess offset in one ledger forced into the available slots of the other, until the shared region runs one consistent update — now read as the transfer of momentum, conserved exactly because the geometry is conserved exactly.

Two consequences are worth stating. First, momentum and kinetic energy are not the same quantity wearing different clothes: kinetic energy is the geometry that was delivered to *establish* the offset (Paper 6.5), and momentum is the standing offset itself. Potential energy, by contrast, is unresolved geometric tension that has not yet been consolidated into an offset — the ledger's record of a configuration's position in a gradient, an accounting that belongs to Paper 10; kinetic and potential energy are the same delivered geometry in two ledger states, consolidated and unconsolidated; they are related but distinct, one the cost of cutting and the other the cut. The framework keeps them distinct for the same reason standard physics does, but grounds the distinction in what each physically is rather than in their differing formulae. Second, the direction of momentum is intrinsic, not relational: the offset points the way it was cut, an absolute fact about the configuration's ledger, which is why momentum is a vector that adds and cancels by direction in collisions — directional geometries pointing oppositely can settle against each other at a junction, exactly as opposed value-surpluses settle in the bonding mechanics of Paper 3.

4. Geometric Consolidation and the Persistence of the Baseline Scan

Figure 3. Geometric Consolidation: Force Writes, Consolidation Saves, the Offset Is Held for Free

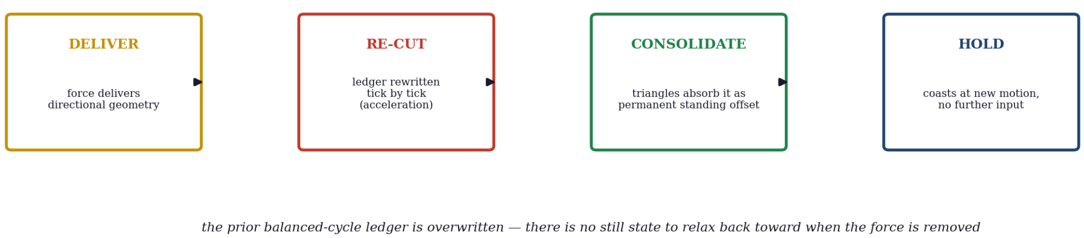

Figure 3. The save-cycle: creation geometry is delivered, the manifold scan is re-cut tick by tick, the configuration's triangles consolidate it as a permanent directional offset, and the new heading is then carried for free — the prior ledger overwritten, with no zero-motion state to relax back toward.

To understand how a configuration retains an updated velocity, the framework discards the classical idea of static rest at the outset, because that idea is not available here. As established in Papers 2, 4, and 6, absolute rest is a structural impossibility (Papers 4, 6): all configurations originate from the photon baseline, the continuous, un-tilted processing rate of the medium, and matter is a configuration that has stepped away from that baseline by translating its processing cycles into structural loops. Matter is not a passive object sitting in an empty metric; it is an active multi-junction composite, and every configuration in the universe is therefore always travelling along some directional vector dictated by its current ledger state. There is no still thing to be set moving; there is only a moving thing whose direction is to be re-cut.

Acceleration, in that light, is not the waking of a passive object from rest. It is the injection of creation geometry into an already-moving circuit. As the impact events of the chemical rocket and the active conduction pumping of the magnetic railway both demonstrate, applying a force introduces raw directional creation geometry directly into the physical universe — genuinely new geometry, not a rearrangement of old. That incoming geometry forces an immediate local overload at the boundary junctions and demands a tick-by-tick re-cutting of the configuration's manifold scan, swinging its standing direction toward the delivered vector.

Geometric consolidation is the mechanism by which this newly added creation geometry becomes a permanent structural feature of the ledger. Because a configuration must continuously resolve its values under the Per-Triangle Rule to hold its closed identity (Paper 5), it cannot simply disregard the influx; it must structurally absorb it. Consolidation is not an ad-hoc compression rule introduced to keep velocity constant after the force stops; it is a stability mandate the Closure Principle enforces on every tick. If the delivered geometry were left loose, sitting in the ledger as un-anchored displacement rather than folded into the directional offset loop, the continuously running identity scan would find a core triplet that no longer locks, and the configuration would fragment, destabilizing into an open, searching waypoint state (States 2 through 8). Consolidation is therefore mandatory hardware maintenance: the configuration packs the delivered geometry into its own standing cycle precisely in order to keep its lock, because the alternative to consolidating is not coasting differently but ceasing to be a closed configuration at all. Consolidation is that absorption made permanent: the incoming creation geometry overwrites the previous configuration values, alters the step size, and forces the continuous scan cycle onto a new heading. When the external delivery stops, the configuration does not snap back or slow toward a halt, because there is no zero-motion state to return to — the prior ledger state no longer exists, having been overwritten (the mechanical answer to the relaxation objection of Paper 4, Section 4). This is the decisive break with the classical picture: classical mechanics imagines a body relaxing toward rest when the force is removed; the framework has no rest for it to relax toward, so it simply continues on its new heading.

The configuration journeys onward at its newly acquired velocity for free, because executing a consolidated ledger entry costs no additional geometry — holding a heading is always free (Paper 6). The new momentum is the permanent localized record of that accumulated creation geometry, held within the rigid, bolted-in atomic structure. (As a configuration is driven toward c the cost of each further re-cut rises, the accounting of which is Paper 8; the slowing of its internal scan under that load is Paper 13.) To alter the path again, or to reverse it, the configuration cannot act passively: an external process or an opposing thruster must introduce new creation geometry to actively re-cut the manifold scan in another direction. There is no default brake. Until a new impact occurs, the consolidated ledger stays locked and the medium continuously honors the rewritten vector. Where the prior section's standing offset was the held quantity, consolidation is the act of writing becoming permanent, and the two regimes — re-cutting at a cost, holding for free — meet exactly at the moment the delivered geometry crystallizes into the standing ledger.

This is also why changing direction requires a side thruster and cannot be done by pushing harder on the main engine — a fact classical mechanics records but does not explain at the level of structure. Firing the main engine adds creation geometry along the existing heading, deepening the offset already written; it cannot pivot the heading, because it injects no asymmetry. To turn, the configuration must take in asymmetric geometry: a side thruster injects creation geometry into a junction cluster at one edge of the body, creating a localized ledger mismatch in which the thrust side is forced to rewrite its directional offset while the rest of the body still runs the old straight-line offset. The body turns because the junctions must resolve that conflict, compressing and pivoting until the whole composite is brought back into a single unified heading. Steering is not changing the vehicle's mind; it is forcing a coordinate re-alignment of the physical hardware at the boundary junctions, the same conduction-pump propagation of Section 2 driven asymmetrically rather than along the axis of travel.

5. Three Demonstrations of the Cycle

Figure 4. Two Engines: Internal Impact-Generation versus External Conduction-Pumping

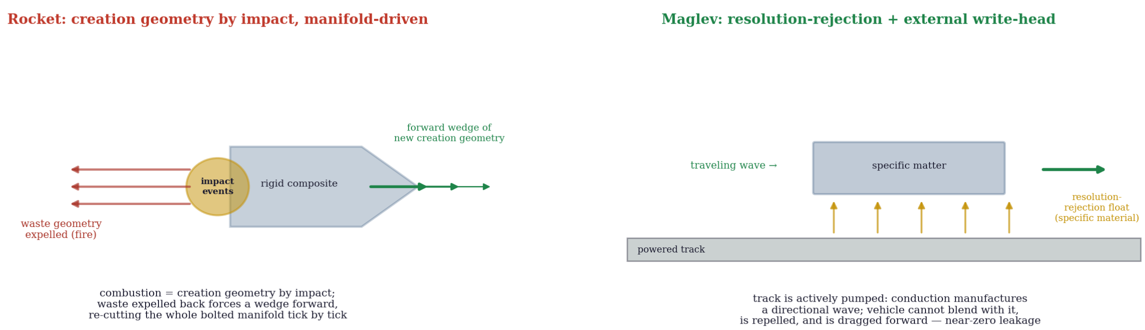

Figure 4. Two ways to deliver directional geometry: the rocket generates it internally by creation events and expels waste to drive a forward wedge into its own matter; the maglev projects a traveling wave into the floating vehicle as an external write-head, consolidating it with near-zero leakage.

The mechanics of Sections 2 through 4 are most clearly seen in two macroscopic machines whose whole purpose is to deliver directional geometry into a configuration. They differ in how they generate and deliver it, and the contrast illuminates the cycle.

5.1 The Chemical Rocket: Internal Generation, Impact Events, and Manifold Driving

A chemical rocket is a macroscopic configuration assembly that accelerates by generating directional geometry internally through localized creation events. Classical mechanics explains rocketry by Newton's third law, attributing the forward push to a passive conservation of momentum as mass is thrown backward across empty space; the standard model tracks this as a macroscopic kinetic exchange between independent particle groups. Both accounts are external and descriptive — they treat the rocket and the exhaust as passive masses interacting across empty coordinates, with no localized mechanism for how the vehicle's internal structure is altered.

In the framework, the engine block is a specialized interaction surface that catalyzes and orients high-frequency creation-geometry events. The combustion of fuel is not mere thermal expansion; it is the rapid, forced disruption of complex multi-junction atomic bonds. When those bonds cross their local closure thresholds, they reorganize under the Per-Triangle Rule, and that violent rearrangement is a cascade of impact events in the localized medium — combustion is, in the framework's terms, creation geometry by impact, the same creation mechanism Paper 4 established, here driven by the engine on purpose. These impact events produce raw, uncompensated creation geometry. Because the engine block is shaped as an asymmetric nozzle, it cannot contain the volatile release locally; it expels the disordered, multi-directional waste geometry rearward as exhaust.

It is worth being exact here about where that geometry comes from, because it is the paper's strongest claim and the one most easily mistaken for magic. The combustion does not merely shuffle pre-existing geometry from one place to another; in breaking the fuel's locked junctions it forces the ledger into a conflict that local transfer alone cannot settle, and to keep its books balanced on the tick the medium generates the precise quantity of new coordinate geometry the conflict demands. This is Strict Generative Accounting, and it obeys a strict trigger law: new geometry is never available for the wishing, and appears only as the mandatory cost of resolving a destabilized junction. The conservation law of this framework is therefore not the classical one, in which a fixed total is endlessly reshuffled; it is the stronger statement that the ledger always balances, where balancing a high-torque conflict requires writing new geometry rather than only moving old. Generation does not break conservation — it is the rule the medium uses to enforce it under load. The universe does not balloon away under this, because the events that trigger generation are specific and local, each tied to an actual structural conflict, a controlled step-by-step process rather than a free expansion. (The summation of these local generative events, together with the outward deposit of unresolved scan workload by the searching states of open space, is the mechanical engine of cosmic expansion, built in Paper 14 and seeded here only as a pointer.)

5.1.1 The Boundary of Generation: the Junction Saturation Threshold

Generation is the strictest expression of conservation under load, not a relaxation of it, and the trigger law makes this exact. The medium does not generate geometry by preference; it generates only when relocation has become mathematically impossible on the tick, and under all ordinary conditions it does not generate at all. A match burning, a spring relaxing, a bond rearranging — these resolve by pure spatial transfer, redistributing existing offsets, because their local coordinate load stays below the medium's native processing ceiling. Generation is forbidden there. The overwhelming majority of all physical change, including most of what looks violent, is transfer.

The trigger is the **Junction Saturation Threshold**, and it behaves like a fuse rather than a dial: it does nothing until the instantaneous local data density — set by the input torque T driving the conflict and the magnitude of the multi-junction coordinate mismatch Delta-x — exceeds what the existing coordinates can clear by relocation in a single tick, and then it must act. Below the threshold the junction discharges by moving geometry; at or above it, relocation cannot close the books, and the Per-Triangle Rule leaves the medium one lawful move — write the exact coordinate geometry the conflict requires, or suffer structural collapse. The threshold is what separates a chemical rearrangement from a combustion-grade rupture: not the chemistry, but whether the conflict's data density overflows what the local coordinates can process by motion alone.

The yield is caged, not open. The volume generated per destabilized junction is fixed by a deterministic, non-linear law, in dimensional form:

> **Delta-G = f(T, Delta-x)**

The function is bounded, not open. Below the threshold the yield is exactly zero; above it the generated volume is capped, in the schematic form 0 <= Delta-G <= kappa *T* Delta-x, where kappa is a geometric factor fixed by the three-axis combinatorics. The coefficients and the exact non-linear shape are deferred to Paper 14, but the bracket is fixed here: zero below threshold, and a hard ceiling proportional to the product of torque and mismatch above it. There is no regime in which a small input yields a large generation.

where Delta-G is the generated coordinate volume, T the driving torque, and Delta-x the mismatch magnitude. This is no blank cheque: f demands a large, destructive input — high torque and severe structural mismatch — to yield even a small, precise volume, so the medium pays the maximum and receives the minimum, the opposite of free energy. The generation is local, instantaneous, and balanced on the very tick of the conflict; the destabilized junction pays a mandatory, exact coordinate price to avoid collapse, and the books close immediately. The closed form of f, the precise location of the threshold, and the dimensional coefficients are fixed by the metric machinery of Paper 14 and assigned there as a locked debt. What this paper fixes is the law's shape and its lock: generation only above the threshold, never freely, and a bounded deterministic yield demanding destructive input for a precise return.

Seed for Paper 14. A local burst is bounded and locked to its conflict, but the global metric cannot delete the written volume once the interaction resolves. The cumulative residual from all such events, with the outward scan-workload deposit of the open-space searching states, is handed to Paper 14 as the mechanical seed of cosmic expansion. The summation to a global effect is a pointer only; this paper claims the local trigger and yield, not the cosmological integral.

Under the strict conservation of the medium, the total geometric pool must stay balanced on every tick. The forced expulsion of waste geometry backward therefore acts as a deterministic driver: it forces a massive, concentrated wedge of brand-new, highly directional creation geometry forward — not merely against the vehicle, but into the entire manifold structure of the engine block, and through it into the physical universe in that direction. The incoming forward geometry floods the interaction surfaces of the composite matter and triggers an immediate local geometric overload at the boundary junctions. The incoming triangles force a continuous, tick-by-tick re-cutting of the configuration's manifold scan. Because the atomic configurations are rigidly bound together as a composite, this ledger update cannot break the assembly apart; instead it is driven through the entire manifold, overriding the old baseline and writing a new, progressive directional offset into the whole vehicle on every tick.

The rocket accelerates because the creation geometry generated by the engine's internal impact events continuously re-cuts the manifold scan in one explicit direction. Once combustion ceases and the creation factory turns off, the accumulated directional creation geometry is structurally consolidated into the standing ledger (Section 4), preserving the new velocity as a permanent, self-sustaining momentum vector. The forward wedge was never a push across a void; it was the universe being given new geometry in one direction, and the vehicle being rewritten to carry it.

5.2 The Magnetic Railway: The External Write-Head and Material Resolution-Rejection

The magnetic railway is the inverse mechanical vector to the rocket: an externally driven system in which the vehicle generates no internal propulsion geometry but acts as a structured composite moving through an external geometry pipeline. Where standard field theories describe magnetic propulsion through abstract, frame-dependent force lines in a vacuum, the framework reads the railway as a macroscopic demonstration of material-specific ledger manipulation and active metric driving. It works through two mechanisms, both grounded in the composition of its materials.

Levitation as static boundary exclusion. The framework reads every magnetic phenomenon as a geometric ledger interaction, with the standard field picture as its statistical shadow (Appendix C); levitation is the static case. What this paper invokes is only the structural account — the rejection of a coordinate gradient by a configuration that cannot absorb it; the full mapping from ledger interactions to electromagnetism is a later concern of the series and is not relied on here. It is not an ambient field reaching out to hold the vehicle but a boundary exclusion between configurations whose geometric tilts cannot share the same coordinate lines. When an ordered configuration is brought near a saturated coordinate gradient — a permanent magnet, or an actively powered electromagnet — the incoming field-tilt meets a structure whose internal junctions are locked into a fixed identity with no capacity to blend with it. The Per-Triangle Rule forbids the two from negotiating a shared junction, the ledger cannot close, and the boundary mismatch lives out as repulsion — the same unresolved condition Papers 3 and 4 identified, where a junction that cannot resolve persists as a standing geometric fact rather than vanishing.

A correction to an overstated earlier claim is owed here, because the audit of this section caught it. The framework does *not* require exotic engineered matter for levitation; ordinary diamagnetic materials — pyrolytic graphite over a magnet, a superconductor expelling a field — levitate, and any account must allow that. What distinguishes a diamagnet at the hardware layer is not exotic composition nor even structural order as such, but *geometric impedance to tilt*: a diamagnetic material's internal junction geometry cannot absorb an incoming coordinate tilt without violating its own closure, so it rejects the tilt rather than taking it in. Graphite achieves this through an intensely ordered planar arrangement; bismuth and a superconductor achieve the same high impedance through different internal geometries — the common factor is not planar order but the inability to absorb the tilt without breaking closure. Plastic does not levitate because its junction geometry has low impedance to tilt — it neither absorbs strongly nor rejects, presenting no firm boundary. The claim is therefore about impedance to tilt absorption, which covers every diamagnet — graphite, bismuth, superconductor alike — not about a single structural motif.

The repulsion is *vertical*, rather than a random directional slide, for a structural reason. The horizontal coordinate axes around the floating body are balanced symmetrically by the surrounding baseline scan — there is no net horizontal mismatch to clear — so the only axis along which the ledger can discharge its unresolvable load is the vertical gradient of the saturation well. The medium shifts the body's spatial-translation ledger upward, along that gradient, to keep the Closure Principle satisfied at the boundary; the float is the ledger taking the one open path. The vehicle holds its height because the gap is packed with mutually rejecting geometry that the ambient baseline load — the downward geometric drift treated fully as gravity in Paper 10 — cannot close. And the exclusion is sustained only while the gradient is: in an electromagnet system the current is the continuous forced injection of creation geometry into the track, so if the current stops, the tilt the body was rejecting vanishes on the tick, the rejection ends, and the body drops. Levitation is an actively maintained boundary state, not a free standing force.

The write-head description that follows is mechanical, not metaphorical: the railway functions as a literal external source of directional geometry, not an analogy for one.

Propulsion as active conduction pumping. The electromagnetic track is an active external geometry factory that does nothing passively: it requires continuous electrical power forcing active conduction across its coils. That pumping drives conduction through the track's atomic structure and generates a high-frequency traveling wave of highly coherent directional geometry directly into the vehicle's path. The wave is a literal macroscopic write-head. When the actively pumped wave meets the vehicle's specific magnetic material configurations, the composite ledger is forced to process the influx, and because the vehicle's rigid structure cannot resolve or absorb the incoming geometric tilt, it rejects the overlap — and that continuous rejection, with the wave moving forward, drags the vehicle's internal offset forward to match the wave's velocity, re-cutting the vehicle's ledger states on every tick. The track wave overrides the vehicle's existing scan cycle continuously, which is the acceleration.

Because the vehicle floats on a zone of pure material resolution-rejection, there is no mechanical contact to break structure or randomize offsets. In a wheeled vehicle, surface grinding constantly scatters the consolidated geometry out of the standing offset into randomized multi-directional resets — momentum leaking away as heat, the friction cascade of Paper 6.5 running slowly and continuously. On the magnetic railway the delivery is clean: the incoming wave writes directly into the vehicle's ledger with near-zero leakage, making it the most efficient macroscopic consolidation mechanism the framework describes.

5.3 The Deep-Space Probe: Microscopic Thermal Asymmetry as an Accidental Driver

Figure 5. The Deep-Space Probe: Thermal Asymmetry as an Accidental Geometry Factory

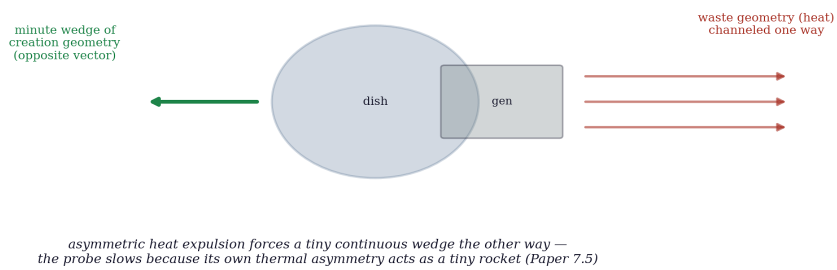

Figure 5. The deep-space probe as an accidental geometry factory: the onboard generator's waste geometry is channeled one way by the antenna dish, forcing a minute continuous wedge of creation geometry the opposite way, which re-cuts the consolidated ledger and slowly decelerates the craft — the framework's reading of the Pioneer anomaly, developed in Paper 7.5.

The probe example is illustrative of the mechanism, not the paper's evidentiary basis for it. A third case shows the same mechanism operating at its finest, most continuous threshold: the unexplained deceleration recorded in deep-space probes and known in the standard literature as the Pioneer anomaly. Standard astrophysics attributes this slight deceleration to asymmetric thermal recoil — infrared photons radiating unevenly off the vehicle's structure and pushing it back.

In the framework, the anomaly is not a passive kinetic bounce but a small, pristine example of a configuration's ledger being re-cut by its interaction with the medium it moves through. Two contributions act together, both of them creation geometry written into the ledger rather than anything lost from it. The first is internal: the vehicle's onboard generator continuously disrupts complex atomic junctions, releasing high-frequency waste geometry as heat, and because the high-gain antenna acts as a directional reflector it channels that waste geometry out along one explicit vector, so that — to balance the local metric on every tick — the asymmetric expulsion forces a minute, continuous wedge of new creation geometry into the rigid structure in the opposite direction, the vehicle acting accidentally as a tiny rocket. The second is external: far from the Sun, the heavily tilted ledger of the craft is continuously interacting with the scan-reduced processing rate of the open-space medium, and that steady rate mismatch re-cuts the consolidated ledger as well. Both write geometry into the same standing offset, subtly and persistently, in the sunward direction; the probe slows because its manifold scan is being rewritten, from within and from without, tick by tick. Which contribution dominates, and how the steady magnitude follows from the medium's processing rate, is the work of the critique companion, Paper 7.5, where the standard thermal-recoil model is examined against the framework's boundary mechanics, the unifying processing-rate mechanism is developed, and the historical dissent the anomaly provoked is taken up.

6. Vector Composition: How Angled Delivery Is Resolved

A configuration moving along one axis and then receiving creation geometry along another composes the two without scrambling its ledger, because the medium is a three-axis continuous framework, and a directional offset is held as a ratio of stepped updates across those three native axes. An acceleration delivered at an angle does not create a new hybrid kind of state; it alters the ratio of stepped updates among the X, Y, and Z axes during the manifold scan. What legacy physics writes as vector addition is, on this reading, an aggregated macroscopic description of discrete, interleaved axis-step ratios — the angle of a resultant is a proportion of axis-steps, not a new direction grafted on.

This composition is not a strict serial queue, one axis finished before the next begins, because a real high-torque event does not wait. It is a simultaneous, load-dependent distribution: the rate at which any single axis updates is a function of the local junction's geometric saturation, so when a burst of creation geometry floods a composite at once, different junctions process the rewrite at different rates according to their orientation and immediate load. This is why a structure under hard, uneven acceleration deforms — a chassis twisting under launch torque is the textbook case — the junctions are not merely moving, they are processing the rewrite at different rates because the delivered geometry is distributed unevenly across them, a differential pressure wave rather than a uniform shove. What structural engineering records as differential loading across a frame, the framework reads as differential processing rate across the junction array. The same mechanism that propagates a re-cut through a body at c (Section 2) distributes an angled or uneven delivery across the body's axes by junction saturation, and the observed twisting is that differential distribution made visible. The steady processing-rate interactions of the companion paper are the slow, continuous version of this same differential rewrite; a launch twist and a deep-space drag are one mechanism at two tempos.

7. Composite Consolidation: A Forward Pointer

One question is raised by the engines and deliberately deferred. A rocket and a maglev vehicle are not single configurations; they are vast composites of bound sub-configurations, and yet each has *one* velocity, *one* momentum, and accelerates as a unit. How do the momenta of many bound sub-configurations consolidate into a single momentum for the whole?

The mechanism must be that the bonds themselves — the locked junctions of Paper 5 — transmit and share delivered geometry across the composite, so that an offset delivered to the back of a rocket propagates through the locked structure until every bound sub-configuration carries a share of the same directional offset, and the composite steps as one. In sketch: a composite is a network of locked junctions (Paper 5); creation geometry delivered to one junction propagates through the locked network via shared offsets, so that every bound sub-configuration receives a share of the same directional offset, and the network enforces a single ledger across the whole, which is why the composite steps as one rather than as a cloud of independent parts. A single momentum for the whole is the signature of a fully consolidated composite, just as a single locked identity is the signature of a closed configuration. But the enumeration of how junctions transmit offset, how much a given bond can carry before it ruptures (the closure threshold of

Paper 5, crossed dynamically), and how a composite's total momentum relates to its constituents' is the accounting of bonded matter, which is Paper 16's assignment. This paper marks the composite consolidation as real and necessary, names its mechanism as offset-sharing across locked junctions, and hands the quantitative treatment forward.

8. Conclusion

Paper 6 left motion as a standing offset and inertia as the cost of changing it; this paper has spent that inheritance. Acceleration is the continuous re-cutting of a configuration's offset, possible only while geometry is delivered, which is why coasting is free and accelerating is not — the force is the delivery the re-cut consumes. Because the re-cutting is a real workload the configuration performs on itself and no co-moving frame shares it, acceleration is absolute and self-detectable, and this is precisely where the hidden offset of Paper 6 surfaces into observability. Momentum is the standing offset regarded as a quantity — accumulated directional geometry, indivisible and quantized, conserved exactly in every interaction because the baseline cannot hold any of it and it can only transfer whole across junctions. And geometric consolidation is the mechanism that binds the two: delivered geometry is structurally absorbed into the standing ledger as a permanent offset, overwriting the prior state so that the configuration continues at its new motion with nothing further supplied. The rocket shows the cycle with an internal geometry factory expelling waste to drive a forward wedge into its own matter; the maglev shows it with an external write-head delivering geometry frictionlessly into a floating, near-lossless vehicle; and the deep-space probe shows it running accidentally and continuously, a craft slowed by its own thermal asymmetry acting as a tiny rocket. One quantity — geometry — passes through delivery, re-cutting, and consolidation, and comes to rest as momentum: the saved file of motion, written by force, held for free. It is conserved under Strict Generative Accounting, which means the books balance on every tick even where balancing a high-torque conflict requires the medium to write new coordinate geometry rather than only move old — generation being the rule that enforces conservation under load, not a breach of it. The composite case, in which many such momenta consolidate into one across locked junctions, is handed to Paper 16; the increase of the re-cutting cost as a configuration is driven toward c is handed to Paper 8. What this paper fixes is the engine of change itself: how a state of motion is rewritten, and how the rewriting stays.

References

This paper is a Tier 1 foundational paper of the *Departure From the Photon Baseline* series and builds on the framework established in the companion papers below.

1. Kadunic, D. (2026). *The 27 Lagrangians of Gravity: A Survey of the Field's Standards of Evaluation* (Paper 1). Zenodo. https://doi.org/10.5281/zenodo.20434286

2. Kadunic, D. (2026). *The Three-Axis Framework and the Nine-State Classification of the 27 Geometric States* (Paper 2). Zenodo. https://doi.org/10.5281/zenodo.20510019

3. Kadunic, D. (2026). *They Assumed It; They Did Not See It: A Critique of Inferred Particles in Standard Physics* (Paper 2.5). Zenodo. https://doi.org/10.5281/zenodo.20518514

4. Kadunic, D. (2026). *The Nine-State Classification with Inside-Outside Structure* (Paper 3). Zenodo. https://doi.org/10.5281/zenodo.20561941

5. Kadunic, D. (2026). *The Constant That Does Not Stay Constant: A Structural Audit of the Fine-Structure Measurement* (Paper 3.5). Zenodo. https://doi.org/10.5281/zenodo.20575611

6. Kadunic, D. (2026). *The Photon as Baseline, Electron-Positron as First Tilt* (Paper 4). Zenodo. https://doi.org/10.5281/zenodo.20633821

7. Kadunic, D. (2026). *The Fragmented Metric — A Structural Deconstruction of Targeted Ruling and the Proliferation of Unfalsifiable Ontologies* (Paper 4.5). Zenodo. https://doi.org/10.5281/zenodo.20634735

8. Kadunic, D. (2026). *The Closure Principle and Configuration Stability* (Paper 5). Zenodo. https://doi.org/10.5281/zenodo.20690435

9. Kadunic, D. (2026). *The Administrative Metric — Effective Field Theory as a Governance Tool for Shifting Rulers* (Paper 5.5). Zenodo. https://doi.org/10.5281/zenodo.20690704

10. Kadunic, D. (2026). *Rest, Uniform Motion, and Inertia as Active Internal Ledger States* (Paper 6). Zenodo. https://doi.org/10.5281/zenodo.20708857

11. Kadunic, D. (2026). *The Passive-Motion Fallacy — Kinetic Energy, the Ghost in the Reference Frame, and the Vindication of the Medium* (Paper 6.5). Zenodo. https://doi.org/10.5281/zenodo.20708957

Paper 6 establishes the standing offset, the balanced and offset cycles, inertia as the workload of re-cutting a cycle, and the topological invariance that makes an offset self-sustaining. Paper 6.5 establishes kinetic energy as delivered geometry and the collision mechanics by which offset transfers whole between ledgers, on which the conservation of momentum in Section 3 rests. Paper 4 establishes energy as added geometry and creation events; Paper 5 the locked junctions through which composite consolidation operates; Paper 4.5 the resolution-rejection that Section 5.2 reads as magnetic levitation. The quantitative law of mass increase toward c is the work of Paper 8; composite momentum is the work of Paper 16; gravity as the ambient baseline load is the work of Paper 10; time dilation is the work of Paper 13; none is pre-empted here.

A. The Bookkeeping of Added Creation Geometry

This appendix redefines the standard terms of motion to show that they are not separate quantities but stages of one process: the introduction, routing, and saving of creation geometry within the absolute manifold. Every standard name is one stage of the same added geometry.

Stage	What the geometry is doing	Framework reading
Injection	Brand-new directional geometry introduced to the universe via internal impact events (rockets) or active conduction pumping (railways)	The input vector — what classical physics calls force or work done: the raw volume of creation geometry meeting an interaction surface
The rewrite	The incoming geometry floods the boundary junctions, a local overload forcing the bolted-in structure to update its heading	The re-cut — what classical physics mislabels inertia: the physical workload of rewriting an already-moving ledger step by step
The total cost	The total volume of creation geometry driven into the manifold to shift its heading	The price of the cut — what standard physics calls kinetic energy: the record of how much geometry was added to the universe during the change (Paper 6.5)

Stage	What the geometry is doing	Framework reading
The saved file	The incoming geometry permanently integrated into the rigid multi-junction bonds under the Per-Triangle Rule	Geometric consolidation — the permanent save: the previous ledger state deleted and overwritten by the new directional vector
Sustained flight	The configuration journeying at its updated velocity for free, because executing a saved entry costs no new geometry	Momentum / coasting — the active baseline scan executing its consolidated code along the updated vector until an opposing impact forces a new re-cut

The point is the single right-hand column: force, inertia, kinetic energy, momentum, and coasting are not five magic properties but the receipt, the workload, the price tag, the saved file, and the continued execution of the creation geometry pumped into the matter. Most of the geometry written into the configuration at injection is relocated from the delivering source — the fuel's broken junction locks, the track's pumped conduction — and where a high-torque conflict cannot be settled by transfer alone, the balance is closed by the medium generating the precise new geometry the conflict demands (Strict Generative Accounting). Either way the books balance locally on every tick; from there the geometry is routed, consolidated, and held. The conservation running through every line is conservation under generation: the ledger always balances, by transfer where it can and by generation where it must.

B. The Absolute Asymmetry of Ledgers

This appendix states the mechanical difference between a configuration holding its current heading and one undergoing a directional change, without any reliance on a privileged frame of empty space — the asymmetry is between executing saved code and overwriting it.

Constant ledger progression (uniform velocity). A configuration moving at a constant rate is executing its already-saved, consolidated ledger instructions. Its rigid composite structure processes its current code uniformly across all internal junctions, so there is no internal traffic jam and no data conflict; the ledger is balanced on every tick. The configuration is travelling absolutely through the medium relative to the photon baseline, but any internal instrument running the same consolidated code sees zero local discrepancy, because it shares the heading. Holding a vector is free and unconflicted, and that is exactly why uniform motion is undetectable from inside — not because the velocity is unreal, but because every co-moving instrument is executing the same saved code.

Active ledger overwriting (acceleration). Acceleration is an absolute, self-detectable disruption because it introduces raw creation geometry that does not match the current ledger. When rocket impact events or active conduction pumping flood the system, they cause an immediate local geometric overload at the interaction boundary — a data traffic jam — forcing a physical, tick-by-tick rewrite of the rigid manifold. Because this re-cutting is an active structural workload happening inside the matter itself, it cannot be differenced away by any choice of reference, since no co-moving reference shares an ongoing rewrite; there is only the saved code, which can be shared, and the act of overwriting it, which cannot. The matter physically feels the overload because its own internal code is being forcibly overridden.

The summary the manual needs: a passenger does not feel steady speed because the whole bolted-together structure is running the same saved code smoothly; a passenger feels acceleration violently because the universe is actively rewriting the code inside the atoms, jamming the boundary junctions while the rewrite is driven through. Velocity is executed; acceleration is written; only the writing is felt.

A reader will ask how a purely geometric, deterministic ledger relates to quantum field theory, which is probabilistic and describes forces through virtual particles. The framework does not need to refute QFT; it reads QFT as a statistical shadow of the ledger, in the way the ideal gas law is a statistical shadow of molecular motion. At the level of the continuous medium the ledger is deterministic — it is counting coordinates and resolving junctions tick by tick — but our instruments are macroscopic and far too slow to track an individual manifold scan, so they record averages over enormous numbers of high-frequency scans. The uncertainty QFT treats as fundamental is, on this reading, our inability to track the specific junction-step of a single scan in real time, not an indeterminacy in the medium itself.

What QFT calls a virtual particle is, in these terms, a discrete coordinate tick — a momentary spike in local processing workload as the background framework recalculates its ledger balance, including the native cycles of the un-looped States 2 through 8 that populate even "empty" space. The framework notes, without rancour, an asymmetry worth marking: QFT permits energy to be borrowed from the vacuum and repaid, treating spontaneous fluctuation as fundamental, while objecting that creation geometry written during a structural event is unaccounted; yet the framework's writing is always triggered by a specific structural interaction and balanced locally on the tick, whereas the vacuum fluctuation is triggered by nothing. The framework's claim is the more conservative one: nothing happens without an interaction, and what looks like acausal fluctuation is the scan jitter of an active medium. QFT's precision is real but is the precision of aggregate statistics — the scattering averages of many repetitions inside shielded apparatus — in the way a casino predicts its yearly take without predicting a single spin; it is the precision of averages over a controlled ensemble, not a tracking of the underlying mechanism. This appendix claims only the bridge, not a derivation: the full reconciliation of the ledger's scan statistics with measured scattering amplitudes is the work of the Tier-2 methodology papers (the collider-inference critique, Paper 18), where the framework's reading of accelerator data is developed, and is not performed here.

These definitions are fixed here to prevent semantic drift across the series; later papers use them in exactly this sense.

Active ledger overwriting. The structural workload of forcibly altering a configuration's self-sustaining directional cycle offset through the systematic injection of un-anchored coordinate displacement at an interaction boundary. (This is acceleration, mechanically.)

Geometric consolidation. The mandatory hardware compression process whereby non-symmetrical, delivered coordinate geometry is integrated into a configuration's active ledger cycle to maintain triplet-lock compliance under the Closure Principle. (Not optional behaviour; a stability mandate.)

Conduction-pump lag. The discrete interval, governed by the hardware processing limit c, required for a ledger overwrite to propagate across adjacent shared junctions within a composite macroscopic assembly. (The reason acceleration is not instantaneous across a body, and the reason a hard-driven structure deforms.)

Strict Generative Accounting. The conservation law of this framework: the ledger balances on every tick, where balancing may require generating the precise quantity of new coordinate geometry a structural conflict demands, never freely and only as the mandatory cost of resolving a destabilized junction.

Generation is the mechanism of conservation under load, not an exception to it. (Replaces the classical fixed-total conservation; the summed output across local conflicts and searching-state deposits is the seed of cosmic expansion, Paper 14.)

Simultaneous asynchronous distribution. The processing of creation geometry across a composite manifold as a parallel, load-dependent rewrite, in which the rate of update on any single axis is a function of the local junction's geometric saturation; in high-torque transitions, differential processing rates across the manifold produce mechanical twisting as boundary junctions resolve incoming geometry at rates set by their orientation and load. (The reason an angled or violent delivery composes correctly and twists a frame.)

The word *geometry* carries three distinct meanings in this framework, and conservation applies to each differently. Fixing the hierarchy here prevents the overloaded term from blurring the conservation claims of the main text.

Sense	What it is	Conservation status
Primitive geometry	The X, Y, O values themselves — three of each per triangle, the non-tunable hardware of Paper 2	Absolute: the count of values is fixed and never changes
Offset geometry	The directional imbalance of those values — the standing offset, held as the ratio of stepped updates across the three native axes; this *is* momentum	Exact at the ledger level: transfers whole between configurations, cannot bleed into or out of the baseline
Creation geometry	New offset geometry introduced during a structural conflict, written under load by Strict Generative Accounting	Generated only above the Junction Saturation Threshold, never freely; the mechanism by which offset geometry is introduced without violating primitive conservation

The three are nested. Primitive geometry is the alphabet — the fixed stock of X, Y, O values that is never added to or subtracted from. Offset geometry is the arrangement — how those values are biased to produce a direction of motion, which is what transfers and conserves as momentum. Creation geometry is the writing of new arrangement under extreme load — the coordinate volume the medium must instantiate to keep the books balanced when a conflict cannot be cleared by rearranging what is already present. When the paper says "geometry is conserved," the precise statement is: primitive geometry is absolutely conserved, offset geometry is exactly conserved in transfer, and creation geometry is the bounded, threshold-gated mechanism that lets the first two stay true under load. Offset geometry is the sense meant wherever this paper speaks of momentum.

To ground offset geometry concretely: because every configuration runs at c and must complete its full identity scan each cycle, offset geometry is the uneven distribution of that fixed processing budget across the three native axes. A configuration looping evenly across X, Y, and Z has no net displacement — it runs at c but loops in place. Tilt it by packing more of the budget's steps onto one axis, and the configuration is physically displaced along that axis per cycle. Offset geometry is exactly that numerical asymmetry in the axis steps — the concrete directional bias written into the configuration's ledger — and momentum is its accumulated, consolidated form.